

1 | Corpus and preprocessing

Project 2 - Biomedical Information Retrieval. Analysis date: 2026-09-30. Scope: Parts I-IX, research questions, Final Challenge and a separate one-page Executive Summary. Optional two-domain comparison omitted.

The original frozen corpus contains 1,000 distinct PMID records with nonempty English-filtered abstracts, retrieved 2026-09-28T14:16:04.129215+00:00. Source: NCBI PubMed ESearch/EFetch, relevance ordering. Only abstracts are analyzed; titles and journals are excluded. English status follows the PubMed language filter, not an independent language detector.

Query: ("GLP-1"[Title/Abstract] OR "glucagon-like peptide-1"[Title/Abstract]) AND hasabstract AND english[Language]

Original SHA-256: 0aeec800f63dc342abc1fe836be4e11adff879f096972482c57e0e7dde0eaf67

Tokenization recognizes English words and internal biomedical hyphens (GLP-1 stays one term). Pure numbers are excluded consistently. A retains standalone punctuation; B drops those tokens; C additionally removes the fixed 81-term stoplist; D additionally applies the project's Porter 1980 implementation. Greek/non-English letters are not analyzed. Structured abstract section labels remain in the text.

condition	documents	tokens	vocabulary	average_tokens_per_document	hapax_terms
A	1000	189100	10732	189.1000	4438
B	1000	161515	10691	161.5150	4434
C	1000	109217	10615	109.2170	4434
D	1000	109217	7815	109.2170	3168

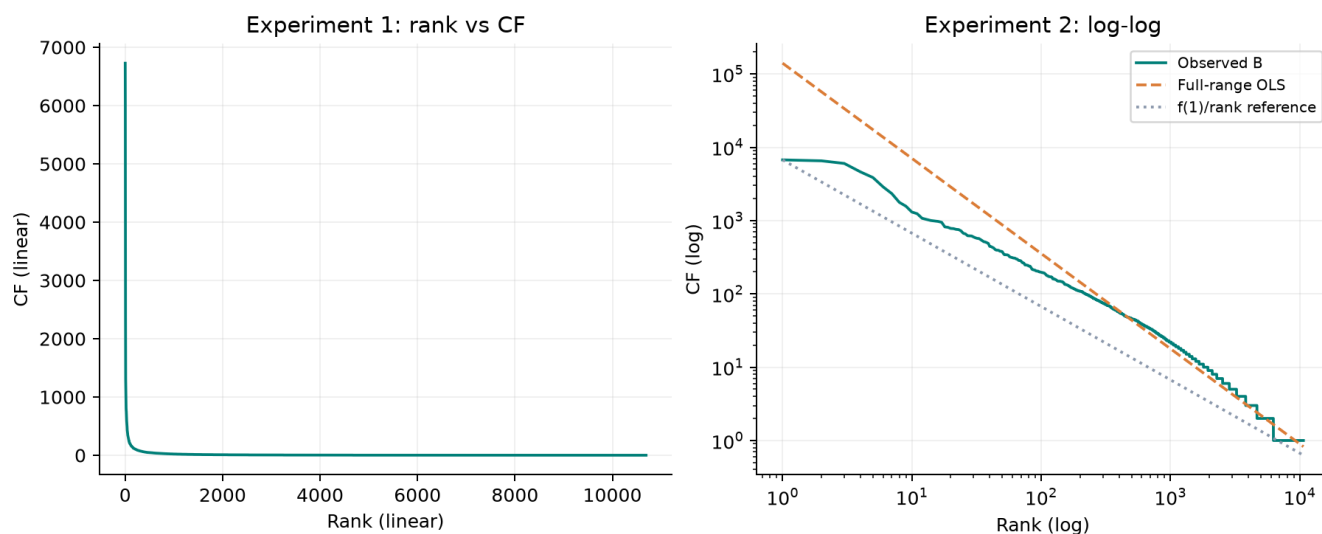
The single-topic relevance-selected snapshot is not a random sample of PubMed. Punctuation handling preserves intraword hyphens. The tokenizer and stoplist are versioned in code; their choices constrain generalization.

2 | Top 50 collection-frequency terms

Condition B: CF sums per-document TF; DF counts documents containing the term. Ranking uses descending CF, then alphabetical tie-breaking.

rank	term	CF	DF
1	the	6727	976
2	and	6530	982
3	of	6015	980
4	in	4617	957
5	glp-1	3863	929
6	to	2876	895
7	a	2331	849
8	with	1772	733
9	for	1570	732
10	is	1307	680
11	that	1240	631
12	on	1074	589
13	receptor	1035	620
14	as	1002	520
15	by	986	520
16	glucagon-like	973	928
17	this	947	655
18	diabetes	819	545
19	effects	809	471
20	peptide-1	781	753
21	are	776	486
22	we	759	475
23	agonists	753	461
24	was	724	351
25	were	671	320
26	have	657	431
27	secretion	621	268
28	these	619	440
29	insulin	616	309
30	treatment	597	394
31	or	582	349
32	glucose	572	307
33	an	569	401
34	weight	551	315
35	type	532	415
36	from	515	351
37	patients	510	281
38	obesity	499	278
39	cells	489	206
40	ras	445	147
41	their	441	292
42	peptide	431	315
43	be	422	315
44	has	404	306
45	may	395	293
46	review	395	322
47	clinical	392	262
48	potential	382	290
49	studies	381	239
50	which	377	295

3 | Rank-frequency and log-log analysis



The linear chart exposes head concentration but compresses the sparse tail. The separate log-log chart reveals the whole range. OLS uses every term; $f(1)/\text{rank}$ is only a $k=1$ reference curve.

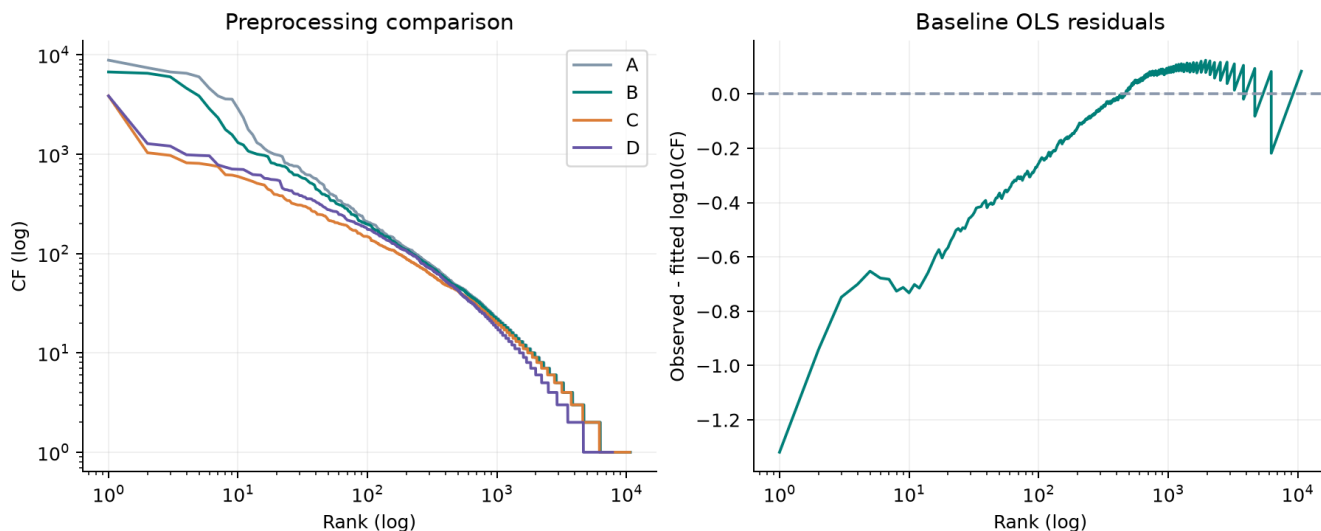
slope	intercept	exponent	r_squared	rmse
-1.2982	5.1468	1.2982	0.9708	0.0975

The full-range B fit is $\log_{10}(\text{CF}) = 5.1468 - 1.2982 \log_{10}(\text{rank})$: $k=1.2982$, $R^2=0.9708$, $\text{RMSE}=0.0975 \log_{10}$ units. The corpus shows a strongly skewed, approximately Zipf-like pattern, but the full-range exponent is above the ideal $k=1$ and residuals are structured. This is descriptive evidence of approximate scaling, not proof of an exact power law.

R^2 is not sufficient evidence: logarithms change the error model, neighboring ranks are dependent, equal frequencies form steps, and the finite corpus truncates the tail. Different distributions may look linear over restricted ranges. Residuals, sensitivity analysis and formal comparisons are needed for a stronger claim. This assignment uses descriptive rank regression, not maximum-likelihood fitting of a frequency probability distribution; their exponents must not be conflated. [Clauset et al., 2009.]

OLS definitions: $x=\log_{10}(\text{rank})$, $y=\log_{10}(\text{CF})$; $\text{slope}=\frac{\sum((x-x_{\text{mean}})(y-y_{\text{mean}}))}{\sum((x-x_{\text{mean}})^2)}$; $\text{intercept}=y_{\text{mean}}-\text{slope}*x_{\text{mean}}$; $k=-\text{slope}$; $R^2=1-\text{SSE}/\text{SST}$; $\text{RMSE}=\sqrt{\text{SSE}/\text{number_of_terms}}$.

4 | Preprocessing and rank regions



condition	vocabulary	exponent	r_squared	rmse
A	10732	1.3142	0.9733	0.0943
B	10691	1.2982	0.9708	0.0975
C	10615	1.2414	0.9610	0.1084
D	7815	1.3487	0.9617	0.1165

B removes 27,585 punctuation occurrences. C removes 52,298 stopwords (32.4%), lowering k from 1.298 to 1.241. D keeps C's token count but reduces vocabulary by 26.4% and raises k to 1.349. Preprocessing alters head mass and tail length, not merely labels.

Top five: A: ,, , the, and, of; B: the, and, of, in, glp-1; C: glp-1, receptor, glucagon-like, diabetes, effects; D: glp-1, effect, receptor, diabet, glucagon-like

region	rank_start	rank_end	exponent	r_squared	rmse
High (first 10%)	1	1070	0.9271	0.9967	0.0229
Middle (next 80%)	1071	9621	1.5462	0.9704	0.0664
Low (last 10%)	9622	10691	-0.0000	undefined	0.0000

Under predetermined rank bands, the head (ranks 1-1070) has $k=0.9271$, $R^2=0.9967$, $RMSE=0.0229$. The middle has $k=1.5462$. The last 10% consists entirely of $CF=1$ terms: slope and $RMSE$ are zero, but R^2 is undefined. The head best matches an ideal inverse-rank relationship: slope closest to -1, small residual error and no constant-frequency degeneracy. Different rank ranges have different variance; R^2 alone is not a fair ranking of regions.

5 | CF versus DF

These 25 terms mix the top ten CF terms with predefined biomedical terms, filled to 25 in CF order. They were not selected to maximize IDF contrast. CF/DF is the average count within documents containing the term.

term	CF	DF	CF_per_DF
the	6727	976	6.8924
and	6530	982	6.6497
of	6015	980	6.1378
in	4617	957	4.8245
glp-1	3863	929	4.1582
to	2876	895	3.2134
a	2331	849	2.7456
with	1772	733	2.4175
for	1570	732	2.1448
is	1307	680	1.9221
glucagon-like	973	928	1.0485
peptide-1	781	753	1.0372
receptor	1035	620	1.6694
agonists	753	461	1.6334
insulin	616	309	1.9935
glucose	572	307	1.8632
diabetes	819	545	1.5028
obesity	499	278	1.7950
semaglutide	160	85	1.8824
liraglutide	204	101	2.0198
exenatide	73	49	1.4898
cardiovascular	342	168	2.0357
patients	510	281	1.8149
treatment	597	394	1.5152
that	1240	631	1.9651

Repetition can yield high CF but relatively low DF. 'fxr' has CF=22, DF=3, averaging 7.33 mentions per containing document. This illustrative example was selected by highest CF/DF among terms with CF>=20 and DF<=10.

DF (or DF/N) measures document coverage. CF measures occurrence volume. Retrieval needs both: CF describes commonness and positional workload; DF describes document spread, postings length and IDF. Neither measure alone establishes clinical importance or relevance.

6 | IDF and a TF-IDF example

$IDF(t) = \ln(N/DF(t))$, $N=1,000$, with natural logarithms and no smoothing. There are no zero-DF vocabulary entries. Changing log base rescales IDF but preserves its ordering. This table includes more than ten terms.

term	CF	DF	IDF
the	6727	976	0.0243
and	6530	982	0.0182
of	6015	980	0.0202
in	4617	957	0.0440
glp-1	3863	929	0.0736
to	2876	895	0.1109
a	2331	849	0.1637
with	1772	733	0.3106
for	1570	732	0.3120
is	1307	680	0.3857
glucagon-like	973	928	0.0747
peptide-1	781	753	0.2837
receptor	1035	620	0.4780
agonists	753	461	0.7744
insulin	616	309	1.1744

'the' has $IDF=0.0243$; 'glp-1' has $IDF=0.0736$; 'semaglutide' has $IDF=2.4651$. Terms found almost everywhere provide little discrimination inside the collection. Topic specificity makes GLP-1 common here even though it might distinguish unrelated domains.

PMID 37192005, chosen as the document with the most 'semaglutide' occurrences, illustrates raw $TF-IDF=TF*IDF$. These are its top ten weights using the same corpus IDF, without length normalization.

term	TF	DF	IDF	TF_IDF
binge-like	4	1	6.9078	27.6310
alcohol	6	14	4.2687	25.6122
drinking	5	6	5.1160	25.5800
semaglutide	9	85	2.4651	22.1859
gaba	3	2	6.2146	18.6438
cea	2	1	6.9078	13.8155
dependence-induced	2	1	6.9078	13.8155
ilc	2	1	6.9078	13.8155
rats	4	57	2.8647	11.4588
aud	2	6	5.1160	10.2320

Zipf describes concentrated collection counts. TF-IDF addresses an associated retrieval problem by downweighting widespread terms while preserving local repetition. High CF does not imply high DF: IDF uses measured DF, not Zipf rank. [Manning et al., Inverse document frequency.]

7 | Information Retrieval implications

Part IX discussion (343 words; required range: 300-500).

Zipf's Law matters to retrieval because it describes an uneven allocation of evidence and storage. In this collection, 10,691 vocabulary entries account for 161,515 token occurrences, but 4,434 terms occur only once. A system designed around a uniform vocabulary would misrepresent both common queries and rare terminology. The useful engineering observation is this imbalance, rather than an assumption that every rank obeys an exact inverse law. [Manning et al., Zipf's law.]

Stopword removal reduces tokens by 32.4% in our experiment. The word 'the' appears in 976 documents, so its presence rarely distinguishes one abstract from another. However, removal is a retrieval policy, not an automatic quality improvement. Negation and phrase structure can matter, especially in biomedical queries. Our fixed teaching stoplist removes 'not'; a production system should evaluate that choice using relevance judgments. We did not measure precision, recall or ranking quality, so fewer tokens cannot be interpreted as better retrieval.

An inverted index stores a dictionary entry for each retained term and a postings list for the documents containing it. Many singleton entries can make dictionary overhead substantial, while widespread terms require long document lists. Repeated occurrences in one document increase CF and positional information without increasing DF. Thus dictionary size, document postings and positional storage have different drivers. Sorted document identifiers allow gap encoding; integer encoders can represent many small gaps compactly. Actual compression depends on ordering and implementation, and this project reports counts rather than measured index bytes. [Manning et al., Index compression.]

TF-IDF combines local repetition with collection-wide rarity. Here 'gip-1' occurs in 929 of 1,000 abstracts, giving IDF 0.074, whereas 'semaglutide' has IDF 2.465. A biologically central topic word can consequently distinguish documents less than a narrower drug name. High CF alone is insufficient to justify downweighting: IDF is determined by DF. Zipf describes global frequency imbalance, while TF-IDF uses document distribution to turn that imbalance into a weighting decision. Rare terms still require scrutiny because spelling variants and noise also receive large IDF values. [Manning et al., Inverse document frequency.]

Final Challenge - why an approximate law remains useful

An approximate empirical law is useful when it guides robust choices without claiming exact predictions. Our head exponent 0.927 and whole-vocabulary exponent 1.298 differ, yet both plots show concentration in a few terms and a sparse tail. That motivates separate treatment of frequent postings and rare dictionary entries, monitoring common queries and using document-aware weights. It does not give an exact storage budget or guarantee relevance. A sound system measures its own corpus, checks residuals and preprocessing sensitivity, and evaluates changes on retrieval tasks. Zipf is a testable design heuristic whose value survives deviations from $k=1$; its limitations show where measurement must replace extrapolation.

8 | Research answers and reproducibility

RQ1 - Approximate Zipf-like structure is visible, particularly in the high-frequency head. The full distribution is not exactly $k=1$; residual curvature and a singleton plateau contradict an exact universal fit.

RQ2 - Baseline B $k=1.2982$; head-only $k=0.9271$. State preprocessing and rank interval with every exponent.

RQ3 - A-D exponents are 1.3142, 1.2982, 1.2414, 1.3487. Stopwords change head mass; stemming merges variants and shortens the tail. A smaller vocabulary does not guarantee better fitting or retrieval.

RQ4 - CF sums TF; DF counts documents with nonzero TF. Repetition increases CF without necessarily changing DF. The 25-term table demonstrates these distinct distributional properties.

RQ5 - Frequency imbalance informs stopword policy, dictionary/postings tradeoffs and document-aware weights. Relevance testing and measured index storage remain separate experiments.

Reproduce: install requirements-report.txt; run `python -m scripts.build_assignment`. The generator checks the original SHA-256 first. Outputs include all terms for A-D, Top 50 tables, region fits, CF/DF/IDF, nonzero document TF and a TF-IDF example. No analysis-time downloads. Existing Word2Vec/search demos are extras, not substitutes for these experiments.

References (the local course PDF is not redistributed):

[1] Course specification, 2026-09-29. [project-2-details-20260929-1.pdf](#) (local document)

[2] Manning, Raghavan & Schutze (2008), Zipf's law.
<https://nlp.stanford.edu/IR-book/html/htmledition/zipfs-law-modeling-the-distribution-of-terms-1.html>

[3] Manning et al., Inverse document frequency.
<https://nlp.stanford.edu/IR-book/html/htmledition/inverse-document-frequency-1.html>

[4] Manning et al., Index compression.
<https://nlp.stanford.edu/IR-book/html/htmledition/index-compression-1.html>

[5] Clauset, Shalizi & Newman (2009), Power-law distributions in empirical data.
<https://arxiv.org/abs/0706.1062>

[6] Porter (1980), An algorithm for suffix stripping. <https://tartarus.org/martin/PorterStemmer/def.txt>

[7] NCBI E-utilities. <https://www.ncbi.nlm.nih.gov/books/NBK25497/>